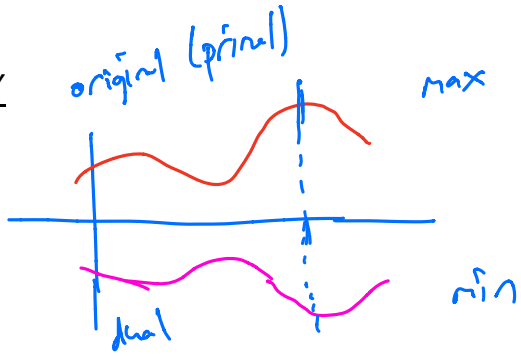
DUALITY



Dual Problem

Each original (or primal) problem is associated with a dual problem. The two problems are closely related, in the sense that the optimal solution of one problem automatically provides the optimal solution to the other. As such, it may be advantageous computationally in some cases to determine the primal solution by solving the dual.

To construct the dual problem, put the primal problem in equation form:

$$\max \text{ or } \min \quad z = \sum_{j=1}^{n} c_j x_j$$

$$s.t. \quad \sum_{j=1}^{n} a_{ij} x_j = b_i \qquad i = 1,2,..,m$$

$$x_j \geq 0 \qquad j = 1,2,..,n$$

m equations
n variables

i. A dual variable is defined for each primal equation (constraint)
ii. A dual constraint is defined for each primal variable.
iii. The const. column coefficients of a primal variable define the LHS coefficients of the dual constraint and its objective coefficient define the RHS.
iv. The objective coefficients of the dual equals the RHS of the primal constraint equations.

	Primal variables					
	x1	x2	.....	xj	.....	xn
Dual variable	c1	c2	.....	cj	.....	cn
y1	a11	a12	.....	a1j	.....	a1n
y2	a21	a22	.....	a2j	.....	a2n
:	:					
ym	am1	am2	.....	amj	.....	amn
				j.th dual const.		

b1
b2
:
bm

Dual obj. Coefficients

$b_1 y_1 + b_2 y_2 + \dots b_m y_m$

$a_{11} y_1 + a_{21} y_2 + \dots + a_{m1} y_m \geq c_1$

Primal Problem Objective	Dual Problem Objective	Constraint Type	Variables sign
Max	Min	≥	Unrestricted
Min	Max	≤	Unrestricted

Ex:

primal

P: $\max \quad z = 5x_1 + 12x_2 + 4x_3$

$s.t. \quad x_1 + 2x_2 + x_3 \le 10$

$2x_1 - x_2 + x_3 = 8$

$x_1, x_2, x_3 \ge 0$

P in equation form:

$\max \quad z = 5x_1 + 12x_2 + 4x_3 + 0x_4$

$s.t \quad y_1 \rightarrow \; x_1 + 2x_2 + x_3 + x_4 = 10$

$y_2 \rightarrow \; 2x_1 - x_2 + x_3 + 0x_4 = 8$

$x_1, x_2, x_3, x_4 \ge 0$

Dual

D: $\min \quad w = 10y_1 + 8y_2$

$s.t \quad y_1 + 2y_2 \ge 5$

$2y_1 - y_2 \ge 12$

$y_1 + y_2 \ge 4$

$y_1 \ge 0$

y_1, y_2 free

$\rightarrow \; y_1 \ge 0$, y_2 free

Ex:

a) P:

$\max \quad z = -5x_1 + 2x_2$

$s.t. \quad -x_1 + x_2 \le -2$

$2x_1 + 3x_2 \le 5$

$x_1, x_2 \ge 0$

$-x_1 + x_2 + x_3 = -2$

P in Equation form:

$\max \quad z = -5x_1 + 2x_2 + 0x_3 + 0x_4$

$s.t \quad y_1 \rightarrow \; x_1 - x_2 - x_3 = 2$

$y_2 \rightarrow \; 2x_1 + 3x_2 + x_4 = 5$

$x_1, x_2, x_3, x_4 \ge 0$

D: $\min \quad w = 2y_1 + 5y_2$

$s.t \quad y_1 + 2y_2 \ge -5$

$-y_1 + 3y_2 \ge 2$

$-y_1 \ge 0$ $\rightarrow \; y_1 \le 0$

$y_2 \ge 0$ $\rightarrow \; y_2 \ge 0$

y_1, y_2: free

b) P:

$\min \quad 6x_1 + 3x_2$

$s.t. \quad 6x_1 - 3x_2 + x_3 \ge 2$

$3x_1 + 4x_2 + x_3 \ge 5$

$x_1, x_2, x_3 \ge 0$

P in Equation form:

$\min \quad z = 6x_1 + 3x_2 + 0x_3 + 0x_4 + 0x_5$

$s.t \quad y_1 \rightarrow \; 6x_1 - 3x_2 + x_3 - x_4 = 2$

$y_2 \rightarrow \; 3x_1 + 4x_2 + x_3 - x_5 = 5$

$x_1, x_2, x_3, x_4, x_5 \ge 0$

D: max $w = 2y_1 + 5y_2$

s.t

$6y_1 + 3y_2 \le 6$

$-3y_1 + 4y_2 \le 7$

$y_1 + y_2 \le 0$

$-y_1 \le 0$

$-y_2 \le 0$

y_1, y_2 : free

$\rightarrow$ $y_1 \ge 0$, $y_2 \ge 0$

let $x_1 = x_1^+ - x_1^-$, $x_1^+, x_1^- \ge 0$

$x_2 = x_2^+ - x_2^-$, $x_2^+, x_2^- \ge 0$

c) P:

$$\begin{aligned} \max \quad & z = x_1 + x_2 \\ s.t. \quad & 2x_1 + x_2 = 5 \\ & 3x_1 - x_2 = 6 \\ & x_1, x_2 \ \ free \end{aligned}$$

P in Equation form:

max $z = x_1^+ - x_1^- + x_2^+ - x_2^-$

$y_1 \rightarrow$ $2x_1^+ - 2x_1^- + x_2^+ - x_2^- = 5$

$y_2 \rightarrow$ $3x_1^+ - 3x_1^- - x_2^+ + x_2^- = 6$

$x_1^+, x_1^-, x_2^+, x_2^- \ge 0$

D: min $w = 5y_1 + 6y_2$

s.t

$2y_1 + 3y_2 \ge 1$

$-2y_1 - 3y_2 \ge -1$

$y_1 - y_2 \ge 1$

$-y_1 + y_2 \ge -1$

y_1, y_2 : free

$\rightarrow$ min $w = 5y_1 + 6y_2$

$2y_1 + 3y_2 = 1$

$y_1 - y_2 = 1$

y_1, y_2 = free

- A free primal variable always corresponds to an equality dual constraint.
- A primal equation produces an unrestricted (free) dual variable.

Max. Problem		Min Problem
Constraints		Variables
≥	↔	≤ 0
≤	↔	≥0
=	↔	unrestricted

Variables		Constraints
≥ 0	↔	≥
≤ 0	↔	≤
Unrestricted	↔	=

$C_{1\times n}$ $X_{n\times 1}$ $A_{m\times n}$ $b_{m\times 1}$ $Y_{1\times m}$

n = # cols. m = # rows

In matrix notation:

$z_{1\times 1} = C_{1\times n} \cdot X_{n\times 1}$

$w_{1\times 1} = Y_{1\times m} \cdot b_{m\times 1}$

Primal		Dual	
max	$z = CX$	min	$w = Yb$
s.t.	$AX = b$	s.t.	$YA \geq C$
	$x \geq 0$		Y unrestricted

$A_{m\times n} \cdot X_{n\times 1} = b_{m\times 1}$

$Y_{1\times m} \cdot A_{m\times n} \geq C_{1\times n}$

Weak Duality Theorem:

For any pair of feasible primal and dual solutions (X, Y)

The objective value in the max problem ≤ the objective value in the min. problem

$z \leq w$

$z^* = w^*$

Furthermore, for the optimal pair (X^*, Y^*) the objective values are equal.

$z^* = w^*$

Proof:

$Y \rightarrow$ max. problem

$y \rightarrow$ min. problem

$YA \geq C$

$C \leq YA$

$Ax = b$

$YAx = Yb = w$

$z = Cx \leq YAx$

$z = C \cdot x \leq YAx = Yb = w$

$z \leq w$

Relations Between Primal and Dual Problems

z^* : optimum value
x^* : optimum solution

Find a lower bound:

Ex: Max. $z = 30x_1 + 100x_2$

s.t
$x_1 + x_2 \le 7$ ✓
$4x_1 + 10x_2 \le 40$ ✓
$x_1 \ge 3$ ✓
$x_1, x_2 \ge 0$ ✓

$x_1 = 3, x_2 = 1 \rightarrow z = 190$ → a lower bound

$x_1 = 3, x_2 = 2 \rightarrow z = 290$ → a new lower bound

$x_1 = 5, x_2 = 2 \rightarrow z = 350$ → a new lower bound.

- The z values of any feasible solution is a lower bound (l).
- Ex: $(x_1, x_2) = (5,2)$ ➔ $z = 350 \le z^*$ where z^* is the optimum value.
- $l \le z^* \le u$

Find an upper bound:

- Step 1: Multiply constraint i by factor u_i (Choose the sign of u_i so that all inequalities are ≤ after multiplication.)
- Step 2: Add up all the obtained inequalities into a resultant inequality.
- Step 3: Make the coefficients of the resultant constraint match the objective function. Then the RHS of the resultant constraint is an upper bound of z^*.

Max. $z = 30x_1 + 100x_2$

s.t

					u_i
x_1	+	x_2	≤	7	× 0
$4x_1$	+	$10x_2$	≤	40	× 10
x_1			≥	3	× −10
x_1			≥	0	× −0
		x_2	≥	0	× −0

$\rightarrow 0x_1 + 0x_2 \le 0$
$40x_1 + 100x_2 \le 400$
$-10x_1 \le -30$
$0x_1 \le 0$
$0x_2 \le 0$

$z = 30x_1 + 100x_2 \le 370$

$350 \le z^* \le 370$

$z^* \le 370$

$z^* \le z = 30x_1 + 100x_2 \le 370$, therefore 370 is an upper bound.

Bound Finding Problem:

- In order to match the objective function, we may have many possible combinations of u_i.
- Each combination corresponds to an upper bound.
- Among all the upper bounds, we want to find the minimum one so that the bound is exactly equal to the max z value.
- So, the original problem is changed to a minimum upper bound finding problem which is equivalent to finding best possible multiplication factors for constraints so that the resultant constraint matches the objective function with a minimum RHS.

$$\max\ z = 30x_1 + 100x_2$$

$$\begin{aligned} x_1 + x_2 &\le 7 \quad (u_1) \\ 4x_1 + 10x_2 &\le 40 \quad (u_2) \\ x_1 &\ge 3 \quad (u_3) \\ x_1 &\ge 0 \quad (u_4) \\ x_2 &\ge 0 \quad (u_5) \end{aligned}$$

$u_i \ge 0$

$$\begin{aligned} u_1x_1 + u_1x_2 &\le 7u_1 \\ 4u_2x_1 + 10u_2x_2 &\le 40u_2 \\ -u_3x_1 &\le -3u_3 \\ -u_4x_1 &\le 0 \\ -u_5x_2 &\le 0 \end{aligned}$$

$$(u_1 + 4u_2 - u_3 - u_4)x_1 + (u_1 + 10u_2 - u_5)x_2 \le 7u_1 + 40u_2 - 3u_3$$

- We want to match the coefficients of x_1 and x_2 here with the coefficients of x_1 and x_2 in the objective function.
- That is

$$\begin{aligned} u_1 + 4u_2 - u_3 - u_4 &= 30 \\ u_1 + 10u_2 - u_5 &= 100 \end{aligned}$$

- We also want to find the minimum RHS, which represent the minimum upper bound of z^*. Therefore, for this original problem, we can formulate a corresponding dual problem:

Primal:

$$\max\ z = 30x_1 + 100x_2$$

$$\begin{aligned} x_1 + x_2 &\le 7 \\ 4x_1 + 10x_2 &\le 40 \\ x_1 &\ge 3 \\ x_1 &\ge 0 \\ x_2 &\ge 0 \end{aligned}$$

Dual:

$$\min\ w = 7u_1 + 40u_2 - 3u_3$$

$$\text{s.t.}\quad \begin{aligned} u_1 + 4u_2 - u_3 - u_4 &= 30 \\ u_1 + 10u_2 - u_5 &= 100 \end{aligned}$$

- We want to make some changes so the primal and dual will be in the standard form.

Minimize $w = 7u_1 + 40u_2 - 3u_3$

s.t.
$$\begin{aligned} u_1 \quad &+4u_2 \quad -u_3 \quad -u_4 \quad &&= 30 \\ u_1 \quad &+10u_2 \quad &-u_5 \quad &= 100 \end{aligned}$$

u_4, u_5 look like an excess (surplus) variable. So they can be changed as follows:

$$u_1 + 4u_2 - u_3 \geq 30$$

$$u_1 + 10u_2 \geq 100$$

$$u_1, u_2, u_3 \geq 0$$

- Some interesting pattern reveal the relations between primal and dual:
  - Dual can be considered as the transpose of the primal problem:

Primal:

Maximize $z = \sum_{j=1}^{n} c_j x_j$

s.t. $\sum_{j=1}^{n} a_{ij} x_j \leq b_i \quad i = 1, \cdots, m$

$x_j \geq 0 \quad j = 1, \cdots, n$

Dual:

Minimize $w = \sum_{i=1}^{m} u_i b_i$

s.t. $\sum_{i=1}^{m} a_{ji} u_i \geq c_j \quad j = 1, \cdots, n$

$u_i \geq 0 \quad i = 1, \cdots, m$

Strong Duality Lemma:

- Let $x = [x_1, \cdots, x_n]$ be any feasible solution to the primal.
- Let $u = [u_1, \cdots, u_m]$ be any feasible solution to the dual.
- If (z value for x)=(w value for u), then, x is optimal for the primal and u is optimal for the dual.

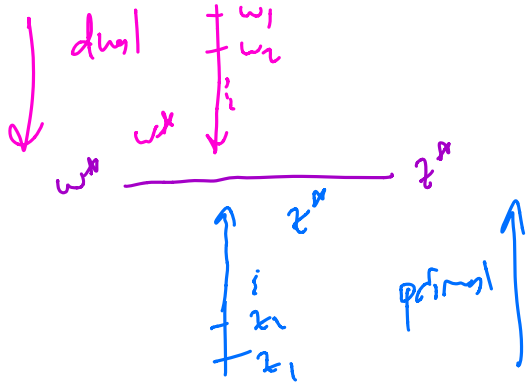
Ex:

$x_1 \quad x_2$

for primal $(3, 2.8)$ is optimal $z^* = 370$

for dual $(0, 10, 10)$ is optimal $w^* = 370$

$u_1 \quad u_2 \quad u_3$



Theorem:

- If the primal is unbounded, then the dual is infeasible.
- If the dual is unbounded, then the primal is infeasible.